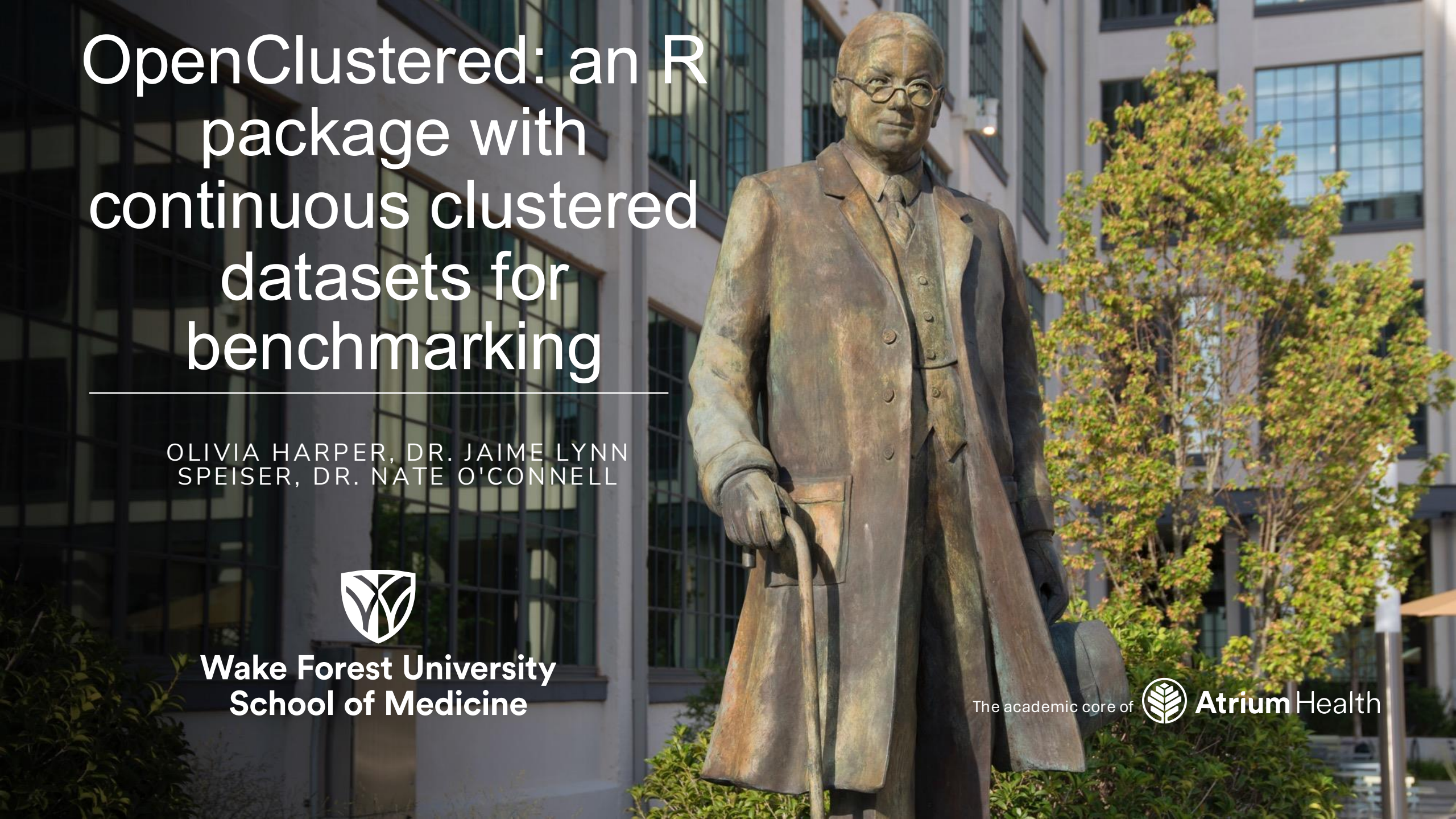
OpenClustered: an R package with continuous clustered datasets for benchmarking

OLIVIA HARPER, DR. JAIME LYNN SPEISER, DR. NATE O'CONNELL

Wake Forest University School of Medicine

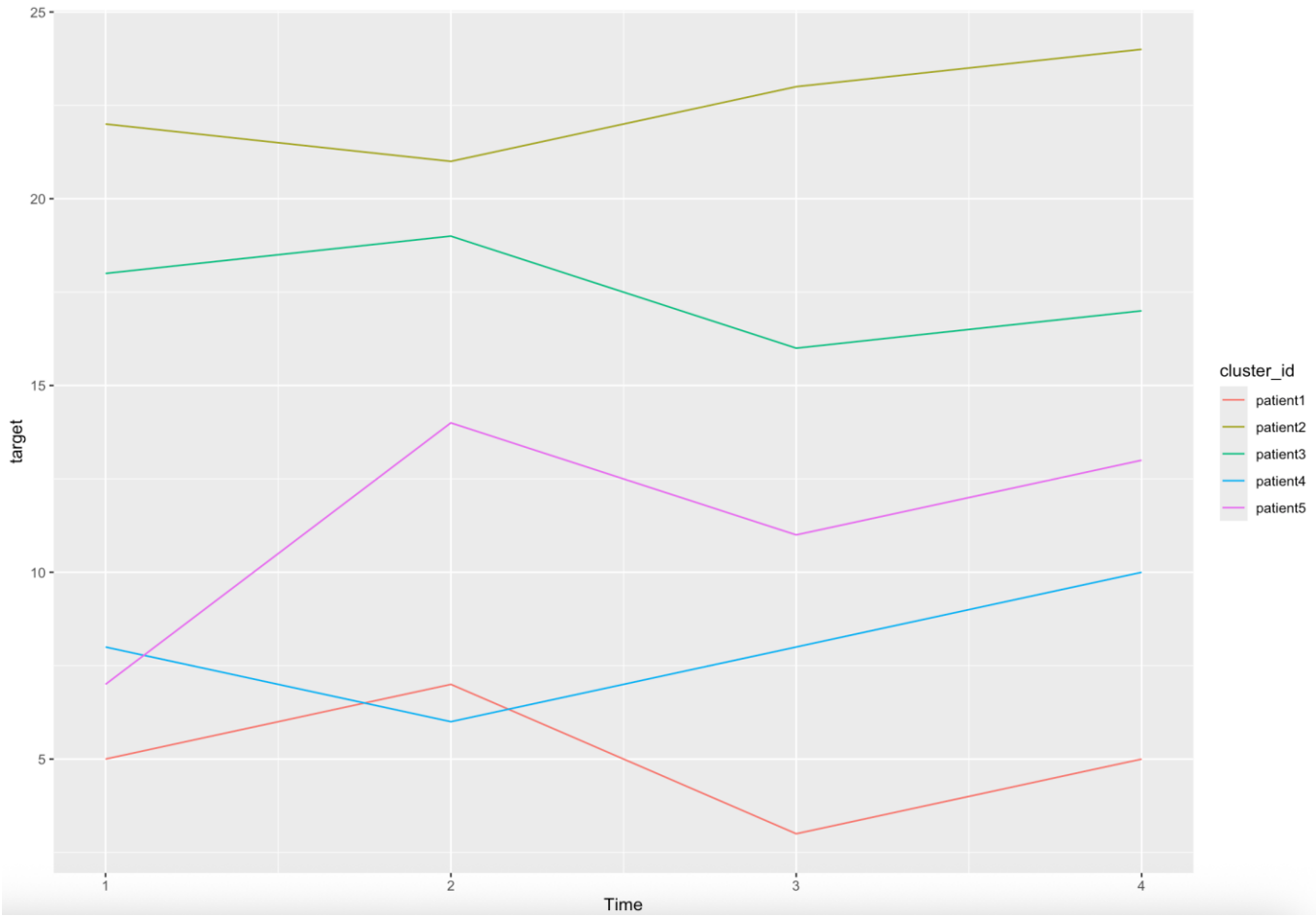
The academic core of Atrium Health



Motivation

What is Clustered Data?

- Grouped/correlated data points
- Violates independence because of grouping
- Requires different analysis methods



Why do we care?

- Clustered data contain characteristics like random slope/ intercepts

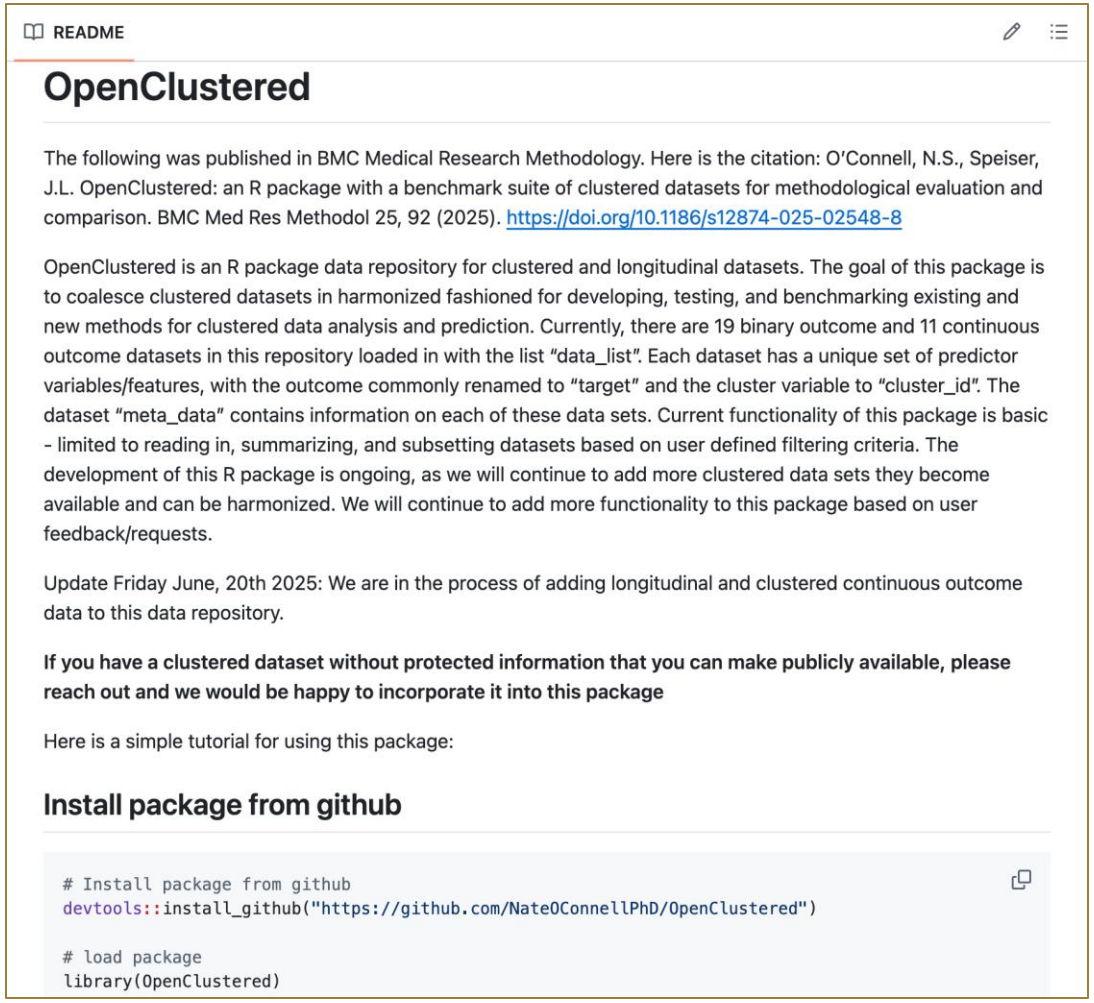
- Few comparison studies to find an optimal method

- This repository provides datasets to evaluate and compare methods

Background and Goal

Background:

- 19 clustered datasets with binary outcomes
- System for finding, preparing, and adding the datasets
- R functions showcasing the metadata
- Housed on GitHub



README

OpenClustered

The following was published in BMC Medical Research Methodology. Here is the citation: O'Connell, N.S., Speiser, J.L. OpenClustered: an R package with a benchmark suite of clustered datasets for methodological evaluation and comparison. BMC Med Res Methodol 25, 92 (2025). https://doi.org/10.1186/s12874-025-02548-8

OpenClustered is an R package data repository for clustered and longitudinal datasets. The goal of this package is to coalesce clustered datasets in harmonized fashioned for developing, testing, and benchmarking existing and new methods for clustered data analysis and prediction. Currently, there are 19 binary outcome and 11 continuous outcome datasets in this repository loaded in with the list "data_list". Each dataset has a unique set of predictor variables/features, with the outcome commonly renamed to "target" and the cluster variable to "cluster_id". The dataset "meta_data" contains information on each of these data sets. Current functionality of this package is basic - limited to reading in, summarizing, and subsetting datasets based on user defined filtering criteria. The development of this R package is ongoing, as we will continue to add more clustered data sets they become available and can be harmonized. We will continue to add more functionality to this package based on user feedback/requests.

Update Friday June, 20th 2025: We are in the process of adding longitudinal and clustered continuous outcome data to this data repository.

If you have a clustered dataset without protected information that you can make publicly available, please reach out and we would be happy to incorporate it into this package

Here is a simple tutorial for using this package:

Install package from github

```
# Install package from github
devtools::install_github("https://github.com/NateOConnellPhD/OpenClustered")

# load package
library(OpenClustered)
```

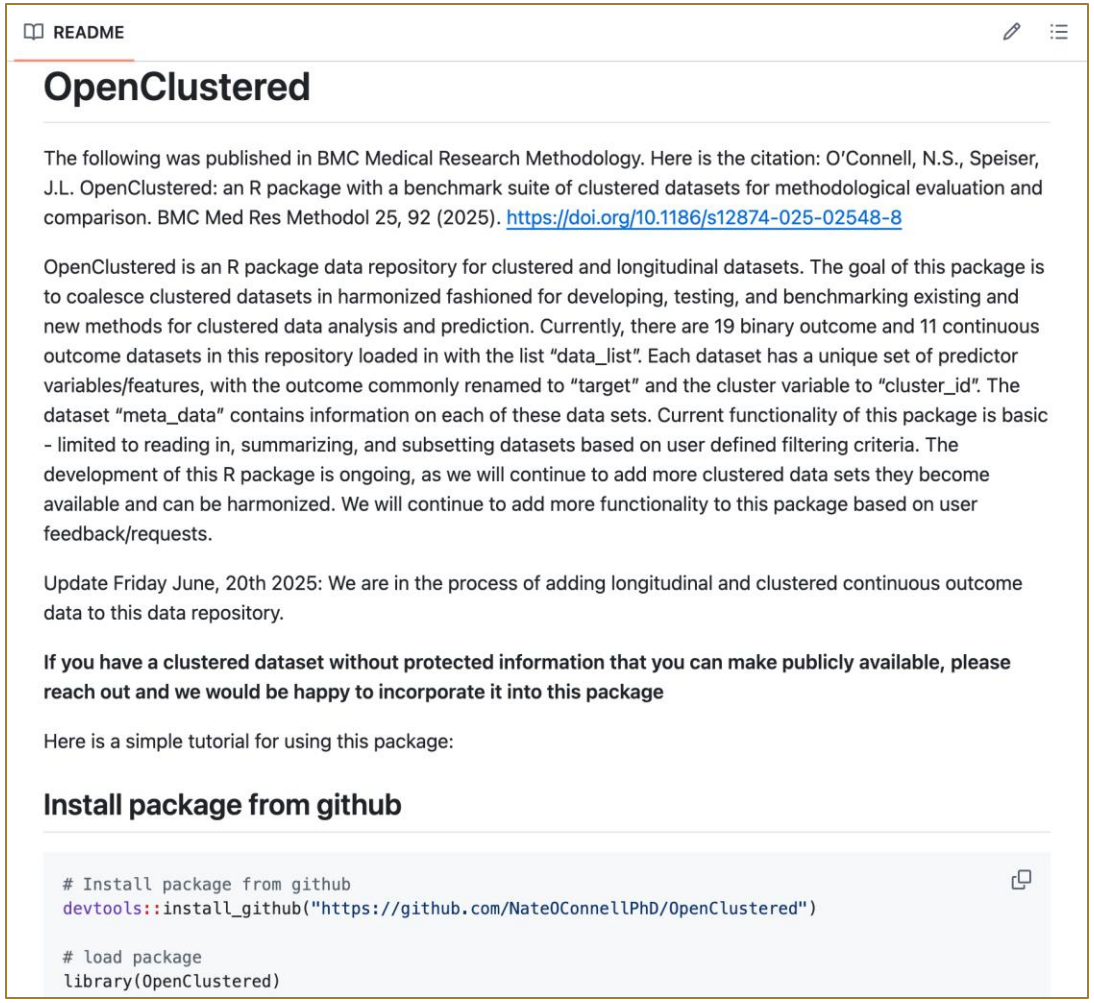
Background and Goal

Background:

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Goal:

- Expand the project to include datasets with continuous outcomes
- Update the functions to be compatible with continuous data



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Methods

Methods

Step 1: Finding Datasets

- Openly sourced:
  - R packages
  - Online tutorials
  - Other open-source repositories
- Search criteria
  - Cluster variable
  - Continuous target outcome
  - 2+ additional variables
- 11 continuous datasets added

Methods

Step 1: Finding Datasets

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  - Online tutorials
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  - Cluster variable
  - Continuous target outcome
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- 11 continuous datasets added

Step 2: Preparing Datasets

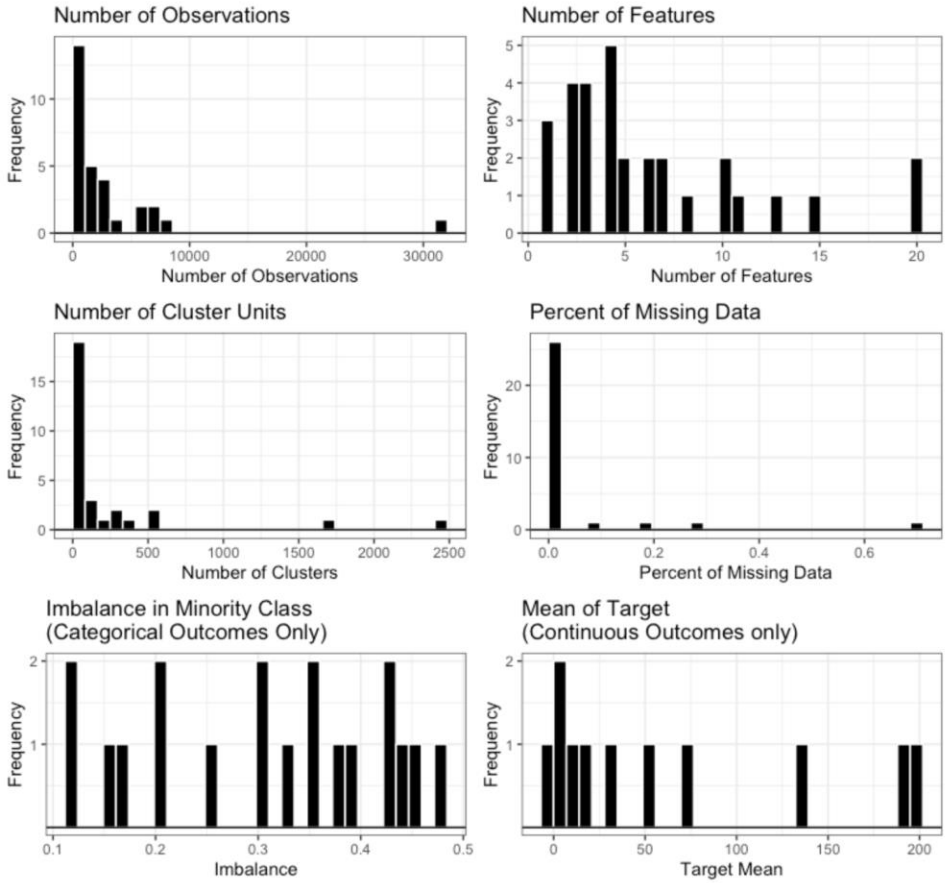
- Standard Formatting
  - Cluster variable --> cluster_id
  - Continuous outcome --> target
- Incorporating the datasets
  - Load_data function
  - Get_meta_data function
- Additions:
  - Target mean
  - Target standard deviation
  - Coefficient of variation



Results

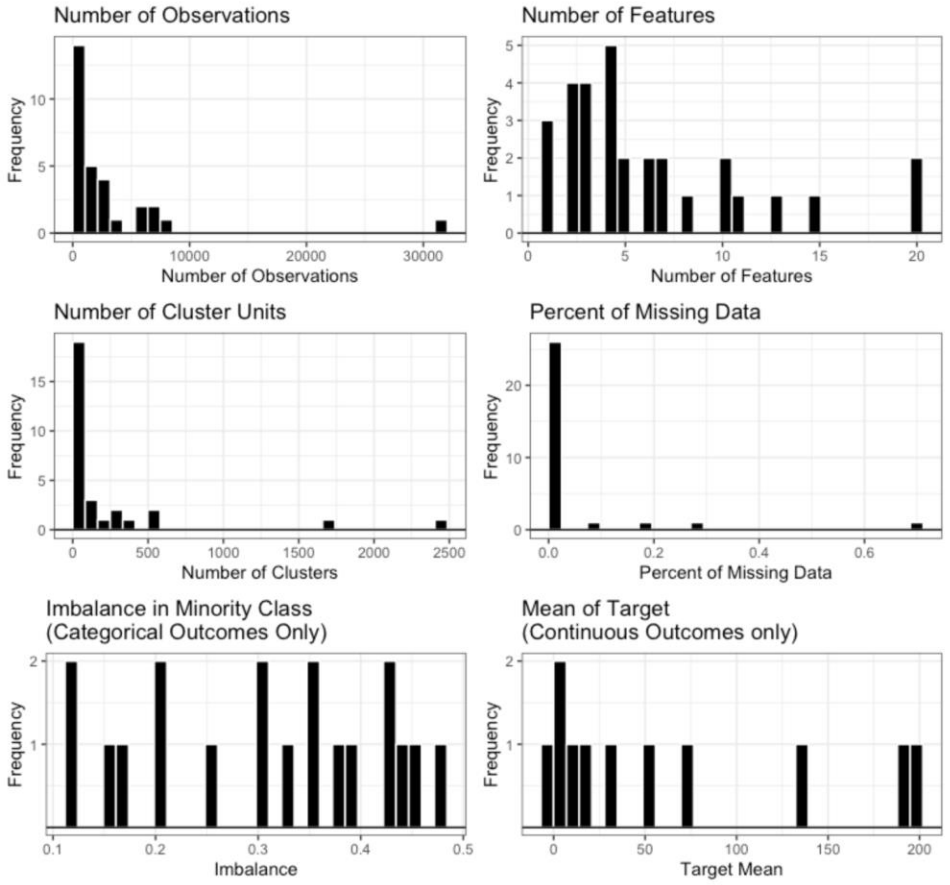
Plot_Meta_Data

Plot_meta_data(allplots = T)



Plot_Meta_Data

Plot_meta_data(allplots = T)



Tab_Meta_Data

tab_meta_data(Task = "both",
~ n_obs + n_features + n_clusters + target_mean + imbalance | task)

	Classification (N=19)	Regression (N=11)	Overall (N=30)
Number of Observations			
Mean (SD)	2060 (2110)	4830 (9190)	3070 (5810)
Median [Min, Max]	1910 [80.0, 8470]	480 [32.0, 31000]	1360 [32.0, 31000]
Number of Features			
Mean (SD)	5.84 (4.92)	6.82 (5.81)	6.20 (5.19)
Median [Min, Max]	4.00 [1.00, 20.0]	4.00 [2.00, 20.0]	4.00 [1.00, 20.0]
Number of Clusters			
Mean (SD)	149 (168)	408 (835)	244 (524)
Median [Min, Max]	60.0 [10.0, 537]	13.0 [6.00, 2410]	55.0 [6.00, 2410]
Continuous Outcome Means			
Mean (SD)	NA (NA)	66.2 (76.0)	66.2 (76.0)
Median [Min, Max]	NA [NA, NA]	29.0 [-0.537, 201]	29.0 [-0.537, 201]
Missing	19 (100%)	0 (0%)	19 (63.3%)
Imbalance			
Mean (SD)	0.309 (0.118)	NA (NA)	0.309 (0.118)
Median [Min, Max]	0.330 [0.120, 0.480]	NA [NA, NA]	0.330 [0.120, 0.480]
Missing	0 (0%)	11 (100%)	11 (36.7%)



Conclusion

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Cluster ------> cluster_id
Outcome -----> target

load_data()

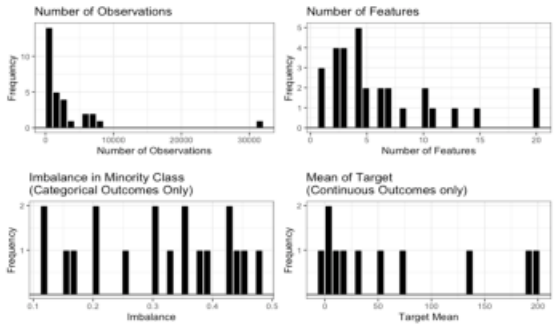
Data_list

[dat1, dat2, dat3, ...]

get_meta_data

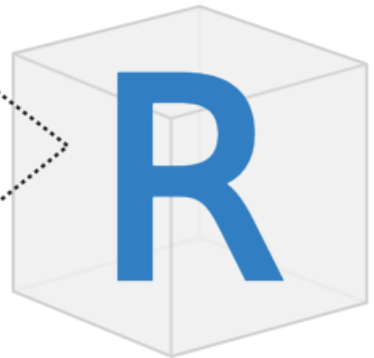
Number of Observations
Number of features
Number of clusters
Number of classes
Task type (classification/regression)
Imbalance (classification)
Target mean (regression)
Target standard deviation (regression)
Target coefficient of variation (regression)

plot_meta_data()



tab_meta_data()

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OpenClustered()

Thank You

